

LAS VEGAS MCCARRAN, NV, USA

WMO: 723860

Lat: 36.072N

Lon: 115.163W

Elev: 665

StdP: 93.59

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23169

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.4	1.9	-18.5	0.8	15.5	-16.7	0.9	13.6	12.3	10.5	11.0	10.4	2.2	200	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	42.8	19.4	41.5	19.1	40.4	18.7	22.6	35.7	21.8	34.8	21.1	34.5	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.8	27.8	17.4	13.5	29.0	16.0	12.3	30.7	69.5	35.7	66.6	34.6	63.9	34.3	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.6	DB	-2.1	44.9	1.7	1.2	-3.3	45.7	-4.3	46.4	-5.3	47.1	-6.6	47.9	(4)
(5)			WB	-5.6	23.8	1.5	1.2	-6.6	24.6	-7.5	25.4	-8.4	26.0	-9.4	26.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.1	9.8	11.7	16.1	19.7	25.0	31.0	34.0	33.0	28.5	21.1	14.1	9.0	(6)	
	DBStd	9.43	3.03	3.54	3.87	4.25	4.39	3.79	2.60	2.39	3.34	3.92	3.82	3.04	(7)	
	HDD10.0	129	41	21	3	1	0	0	0	0	0	0	9	54	(8)	
	HDD18.3	1023	266	185	90	35	4	1	0	0	0	19	134	288	(9)	
	CDD10.0	4192	33	69	192	291	465	630	743	714	556	343	132	24	(10)	
	CDD18.3	2045	0	1	22	76	211	380	485	456	306	103	7	0	(11)	
	CDH23.3	30393	0	8	211	786	2707	6021	8150	7369	4116	973	53	0	(12)	
(13)	CDH26.7	19099	0	0	41	281	1418	3949	5729	4982	2353	343	3	0	(13)	
(14)	Wind	WSAvg	3.6	2.8	3.2	3.9	4.5	4.4	3.9	3.7	3.4	3.2	2.8	2.8	(14)	
(15) Precipitation	PrecAvg	106	13	16	10	4	4	1	9	10	8	7	10	11	(15)	
	PrecMax	250	76	73	122	62	24	12	63	54	52	37	56	53	(16)	
	PrecMin	29	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	52	18	20	18	9	6	3	13	12	12	10	14	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.5	24.7	30.4	34.4	39.5	44.0	44.9	43.0	40.1	34.3	26.9	20.0	(19)	
		MCWB	9.7	11.0	13.1	15.1	18.4	17.9	20.2	20.2	18.8	16.6	12.3	10.2	(20)	
	2%	DB	18.6	22.4	27.6	31.9	37.2	41.8	43.3	41.8	38.3	32.2	24.9	17.9	(21)	
		MCWB	8.8	10.1	12.3	13.9	16.3	17.6	19.7	19.6	18.0	15.5	11.7	8.9	(22)	
	5%	DB	17.2	20.4	25.6	29.9	35.4	40.5	42.0	40.7	37.1	30.3	23.0	16.4	(23)	
		MCWB	8.0	9.3	11.3	13.1	15.5	17.6	19.6	19.2	17.7	14.5	10.9	8.0	(24)	
	10%	DB	15.7	18.5	23.9	27.9	33.5	39.0	40.7	39.4	35.7	28.4	21.1	14.7	(25)	
		MCWB	7.4	8.5	10.6	12.2	14.8	17.2	19.2	18.8	17.4	13.7	10.3	7.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	13.0	14.3	16.2	19.7	22.4	23.9	23.5	22.5	18.6	14.5	11.9	(27)	
		MCDB	16.6	20.4	25.9	32.4	37.7	39.2	38.4	37.4	31.9	28.8	22.2	17.3	(28)	
	2%	WB	10.6	11.5	12.9	14.7	17.7	20.2	22.7	22.4	20.9	17.0	12.8	10.5	(29)	
		MCDB	15.7	18.7	25.1	29.6	34.4	38.3	35.8	35.7	32.0	28.9	22.6	15.7	(30)	
	5%	WB	9.5	10.5	12.0	13.7	16.4	19.1	21.9	21.7	20.1	15.7	11.6	9.3	(31)	
		MCDB	15.0	18.0	23.8	28.1	32.8	37.5	34.7	34.8	31.7	27.3	21.6	14.6	(32)	
	10%	WB	8.4	9.4	11.1	12.7	15.3	18.0	21.3	21.0	19.2	14.8	10.7	8.0	(33)	
		MCDB	13.9	16.9	22.2	26.4	31.8	36.6	35.0	34.1	31.6	26.1	20.0	13.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.3	11.4	11.6	12.1	12.7	12.0	11.8	12.0	11.5	10.7	9.3	(35)	
		MCDBR	11.8	12.9	13.7	14.1	13.8	13.9	13.1	12.8	13.0	13.3	12.7	11.3	(36)	
	5% WB	MCWBR	6.1	6.1	5.9	6.0	5.5	5.4	5.1	5.0	5.3	5.9	6.1	6.0	(37)	
		MCWBR	9.4	10.9	12.9	13.2	13.5	13.4	11.3	11.3	11.5	12.0	11.6	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.264	0.270	0.283	0.305	0.323	0.319	0.359	0.349	0.314	0.291	0.268	0.264	(40)	
		taud	2.618	2.620	2.557	2.474	2.447	2.469	2.443	2.461	2.538	2.591	2.625	2.627	(41)	
	Ebn at Noon	Ebn at Noon	956	987	993	978	958	957	916	922	944	946	943	934	(42)	
		Edh at Noon	72	81	94	108	113	111	113	108	95	82	71	67	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	3.08	4.04	5.54	6.95	7.87	8.54	7.52	6.99	6.18	4.65	3.47	2.73	(44)	
	RadStd	RadStd	0.24	0.35	0.26	0.29	0.39	0.35	0.39	0.35	0.26	0.27	0.14	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.61	N/A	-1.49	N/A	N/A	N/A	N/A	-109	+220	+120	(46)
(47)	Regional (15 neighbors)	+0.44	N/A	-1.60	N/A	N/A	N/A	N/A	N/A	+159	N/A	(47)

Nomenclature: See separate page